

NAME:_____

Math 227C Homework 5

Spring 2026

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1. Consider the predator-prey model with three reactions that we defined in class: $Y_1 \rightarrow 2Y_1$, $Y_1 + Y_2 \rightarrow 2Y_2$, $Y_2 \rightarrow \phi$, with stochastic rate constants $c_1 = 1$, $c_2 = 0.005$, $c_3 = 0.6$.
 - a) Write code to simulate this system stochastically, with initial condition of 50 prey and 100 predators. Write the timecourse both as a function of time and on a phase plot of Y_1 vs Y_2 .
 - b) Simulate the same system deterministically and plot the same graphs in order to compare their behavior.
 - c) Consider lowering the number of predators and prey to the point that the long term behavior of the solution can be qualitatively different from that of the deterministic system. Explain intuitively how this can take place in terms of the predators and prey.
2. Consider the reaction network $A + B \rightarrow C$, $C \rightarrow A + B$ discussed in class.
 - a) Simulate this system deterministically, by setting fixed parameter values, and plot the concentration of A, B, C over time.
 - b) For the same parameter values as in a), simulate this system stochastically, and compare the two solutions.
 - c) Extend this comparison to increasingly high values of the initial conditions, simulated over a fixed time interval $[0, T]$. Do you see a convergence between the stochastic and the deterministic solutions?
 - d) Write down the master equation $\pi'(t) = \pi(t)Q$ for this specific system.
3. Consider the SIR model in reaction form, $S + I \rightarrow 2I$ with rate constant β , and $I \rightarrow R$ with rate constant γ . Here S, I, R stand for susceptible, infected, and recovered individuals.
 - a) Write down the associated deterministic ODE for this system, and confirm that it is the classic SIR model.
 - b) For initial conditions $S(0) = N$ large, $I(0) = \epsilon$ small, and $R(0) = 0$, show that $I'(0) > 0$ if and only if $\beta N / \gamma > 1$. This constant has the name R_0 or reproductive number.
 - c) Set $N = 100$ and $\epsilon = 3$. Simulate the system stochastically for values of R_0 slightly larger than 1, and show by plotting multiple timecourses for each parameter choice how when $R_0 > 1$ an epidemic tends to take place.
 - d) Set $N = 100$ and $\epsilon = 3$. Simulate the system stochastically for values of R_0 smaller than 1, and show by plotting multiple timecourses for each parameter choice how when $R_0 < 1$ an epidemic does not tend to take place.
 - e) Argue why R_0 can be considered to be the average number of people infected by each infectious individual. Hint: calculate the average time that an infected person remains infected, and with what frequency they infect other individuals during that time.